

LESSON 6-6 • SESSION 2 • PRACTICE • FORM A

Testing and Justifying Equivalence

Learning target – I can test whether two expressions are equivalent by substituting values and by reasoning. · 6.EE.A.4

REMEMBER

- Equivalent expressions have the **same value for every** value of the variable.
- **Test** with the same value in both. One value with different results proves they are **not** equivalent.
- Matching values do not prove it. **Justify** with properties, like the Distributive Property.

WORKED EXAMPLE

Are $4(n + 3)$ and $4n + 12$ equivalent?**Step 1** Test $n = 2$: $4(2 + 3) = 20$ and $4 \times 2 + 12 = 20$.**Step 2** One match is not proof, so use a property.**Step 3** Distributive Property: $4(n + 3) = 4n + 4 \times 3 = 4n + 12$.**Answer** Yes, equivalent

WORDS equivalent expressions (*expresiones equivalentes*) · counterexample (*contraejemplo*) · **Distributive Property** (*propiedad distributiva*) · **Commutative Property** (*propiedad conmutativa*)

PRACTICE

show your work · label units

1. Evaluate each expression when $y = 9$. Which two might be equivalent?

a. $2(y + 7) =$ ____

b. $2y + 14 =$ ____

c. $2y + 7 =$ ____

2. **Ticket Formulas.** Booth A uses $6t + 5 + 2t$ for the cost of t tickets. Booth B uses $8t + 5$.

a. Substitute $t = 3$. A: ____ B: ____

b. Substitute $t = 10$. A: ____ B: ____

c. Are they equivalent? How do you know for every t ? _____

d. Which property lets you reorder $6t + 5 + 2t$? _____

3. **One Test Is Enough.** Compare $3(k + 2)$ and $3k + 2$ using $k = 4$. Are they equivalent?

Hint Do the parentheses first in $3(k + 2)$.

Show your work.

Equivalent? _____

4. **Properties.** Look at $2 + (m + 11)$ and $(2 + m) + 11$.

Hint Only the grouping changed.

a. Which property is shown? _____

b. Test both with $m = 6$. _____

5. **Nia's Claim.** Nia says $6 + 5j$ and $11j$ are equivalent because $6 + 5 = 11$. Test $j = 2$ to justify or correct her claim.

Hint Substitute 2 into both expressions.

Show your work.

Values _____

6. **Two Test Values.** Compare $9v + 6$ and $3(3v + 2)$ using $v = 1$ and then $v = 7$. Are they equivalent?

Hint Matching values support it; a property proves it.

Show your work.

Values _____

- 7. Kenji's Claim.** Kenji says $12m - 4m$ and 8 are equivalent because $12 - 4 = 8$. Test $m = 5$ to justify or correct his claim.

Hint Subtracting coefficients leaves a term that still has m .

Show your work.

Values _____

- 8. Three Expressions.** Use $3(k + 4)$, $3k + 12$ and $3k + 4$.

- a. Evaluate all three when $k = 2$. _____
- b. Which two are equivalent? _____
- c. Justify with a property, not a test value. _____

FIND THE ERROR

explain the mistake, then fix it

- 9. Dmitri's Work.** Dmitri tested $4p + 6$ and $10p$ using $p = 1$. Both gave 10, so he said they are always equivalent. Explain his mistake. Test $p = 2$.

TEST PRACTICE

choose the best answer

- 10.** $2(x + 5)$ is equivalent to –
- A $2x + 5$

B $2x + 10$

C $x + 10$

D $7x$
- 11.** If two expressions give different values for one input, they are –
- A equivalent

B not equivalent

C sometimes equivalent

D like terms

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

65 Q	25 P	40 Y	20 Z	18 P	19 E	16 R	85 O	29 R	69 T
#1c	#2a	#2b	#3	#4b	#5	#6	#7		

CHECK YOURSELF

☐ I can do this on my own
 ☐ I need a little help
 ☐ I have a question about #____